

Advanced (Habanero)

Q1. Factor the trinomial $x^2 + 5x + 6$.

- (A) $(x + 2)(x + 3)$
- (B) $x(x + 5) + 6$
- (C) $(x + 1)(x + 4)$
- (D) $(x + 3)(x + 2)$
- (E) $x(x + 6) + 5$

Q2. Factor the trinomial $x^2 - 7x + 12$.

- (A) $x(x - 7) + 12$
- (B) $x(x - 3) - 4$
- (C) $x(x - 4) - 3$
- (D) $x(x - 3) + 12$
- (E) $(x - 3)(x - 4)$

Q3. Factor the trinomial $x^2 + 2x - 15$.

- (A) $x(x - 3) + 2x + 6$
- (B) $x(x + 2) + 3x - 15$
- (C) $x(x + 5) - 3x - 15$
- (D) $x(x + 5) - 3x - 15$
- (E) $x(x + 5) - 3x + 15$

Q4. Factor the trinomial $x^2 - 4x - 21$.

- (A) $x(x + 4) + 3x - 21$
- (B) $x(x + 3) - 3x - 21$
- (C) $x(x + 7) - 7x - 21$
- (D) $x(x - 3) + x - 21$
- (E) $x(x - 7) - (x + 3)$

Q5. Factor the trinomial $x^2 + 10x + 24$.

- (A) $x(x - 4) + 2x + 24$
- (B) $x(x + 2) - 3x - 24$
- (C) $x(x + 4) + 3x - 24$
- (D) $x(x + 4) + 6x + 24$
- (E) $x(x + 4) + 6x + 24$

Q6. Factor the trinomial $x^2 - 9x + 20$.

- (A) $x(x + 4) - 3x + 20$
- (B) $x(x + 2) - 3x + 20$
- (C) $x(x + 5) - 5x + 20$
- (D) $x(x - 5) + 5x + 20$
- (E) $x(x - 4)(x - 5)$

Q7. Factor the trinomial $2x^2 - 5x - 3$.

- (A) $2x^2 - 6x + x - 3$
- (B) $2x(x + 1) - 3(x + 1)$
- (C) $x(2x - 1) - 3(2x + 1)$
- (D) $2x^2 - 6x + x - 3$
- (E) $x(2x + 1) - 3(2x - 1)$

Q8. Factor the trinomial $3x^2 + 2x - 5$.

- (A) $3x^2 + 6x - 4x - 5$
- (B) $3x(x + 2) - 1(3x + 5)$
- (C) $3x(x + 2) - 1(3x - 5)$
- (D) $x(3x + 2) - (3x - 5)$
- (E) $x(3x + 1) - (3x - 5)$

Open Ended Questions (FRQ)

Q9. Factor the trinomial $4x^2 - 20x + 16$.

Q10. Factor the trinomial $2x^2 + 5x - 3$.

Chef's Tips

- Chef's Tips
- Provide necessary space for calculations and answers
- Ensure students to keep their work clean and organized